

OCS IITH Room Booking System

Task Document

Deadline: 9th May, 11:59 PM

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Document Purpose: This document defines the complete task specification for building a secure room booking system for the Office of Career Services (OCS), IIT Hyderabad. The system is intended for booking and managing rooms for OA, interviews, and pre-placement talks, while enforcing authentication, authorization, capacity checks, and time-slot conflict detection.

1. Project Overview

Design and develop a secure room booking system for the Office of Career Services (OCS), IIT Hyderabad. The system will allow authorized users to book rooms for placement-related activities such as online assessments, interviews, and pre-placement talks, while enforcing strict access control, room capacity limits, and time-slot conflict checks.

2. Objective

The main objective of this system is to centralize and simplify room allocation for OCS activities at IIT Hyderabad. The system should ensure that:

- only authorized people can access the system,
- room bookings are made without overlap or conflict,
- room capacity is respected,
- every booking is linked to a valid purpose,
- admins retain full control over user access and room data.

3. Supported Booking Purposes

The system should support bookings only for the following purposes:

1. Online Assessments (OA)
2. Interviews
3. Pre-Placement Talks (PPTs)

4. Authentication and Access Control

The system must use proper authentication and must not allow open registration.

4.1 Rules

- Users cannot sign up by themselves.
- Only an admin can create user accounts.
- Admin can assign access to specific cores or authorized members.
- Each user must log in using admin-provided credentials.
- Admin must be able to revoke access whenever required.

4.2 Recommended Roles

- Admin: full control over users, rooms, and bookings.
- Authorized Core Member: can search and book rooms as per assigned permission.
- Viewer/Read-only User (optional): can only view bookings and schedules.

5. Room Management

IIT Hyderabad has multiple blocks and rooms, such as A Block, B Block, C Block, CSE Block, LHC, and others. The system should be designed to store and manage room data from all such blocks.

5.1 Room Data Fields

Each room entry should contain at least the following information:

- Block name
- Room name or room number
- Seating capacity
- Availability status
- Allowed booking purposes, if any restriction exists
- Notes or special constraints, if applicable

5.2 Dummy Table: Room Capacity Reference

The following table is a dummy/reference table prepared from the room details shared in the image. It can be updated later when the final verified block-wise list is shared.

Room Name	Capacity
A-Class Room 320	80
A-AUDITORIUM	289
A-Class Room 111	70
A-Class Room 112	80
A-Class Room 114	36
A-Class Room 117	84
A-Class Room 118	84
A-Class Room 119	108
A-Class Room 220	40
A-Class Room 221	120
A-LH-1	184
A-LH-2	184
BT/BM-009	24

BT/BM-010	24
BT/BM-118	60
C-LH-10	68
C-LH-2	138
C-LH-3	100
C-LH-4	60
C-LH-5	60
C-LH-6	60
C-LH-7	70
C-LH-9	66
CSE-LH-01	70
CSE-LH-02	70
CSE-LH-03	70
CY-LH-1	30
CY-LH-2	40
CY-LH-3	90
EE-004(GF)	80
EE-20 (SF)	60
LHC-01	72
LHC-02	72
LHC-03	120
LHC-04	200
LHC-05	800
LHC-06	320
LHC-07	200
LHC-08	120
LHC-09	72
LHC-10	72
LHC-11	120
LHC-12	200
LHC-13	320
LHC-14	200
LHC-15	120
MA-01	56
MA-02	56
MA-114	30
MSME-LH-1	36
MSME-LH-2	60
MSME-LH-3	106
PH-1	80
PH-2	60
PH-3	50

5.3 Future Expansion

The room database should be flexible enough to accept additional blocks and rooms later when the full list is shared.

6. Booking Requirements

The booking module should allow authorized users to reserve rooms based on purpose, capacity, and time availability.

6.1 Booking Inputs

A booking request should include:

- date
- start time
- end time
- purpose of booking
- expected number of participants
- selected block and room

6.2 Booking Constraints

The system must enforce the following rules:

- A room cannot be double-booked for overlapping time slots.
- The selected room must have capacity greater than or equal to the expected participant count.
- Only allowed purposes can be selected.
- A booking must be made only by an authenticated and authorized user.
- Admin may cancel or override bookings when necessary.

7. User Workflow

A typical booking flow should work as follows:

4. Admin creates user accounts and grants access.
5. Authorized user logs in.
6. User selects purpose, date, time, and expected capacity.
7. System shows eligible rooms based on block, availability, and capacity.
8. User chooses a room and submits the booking request.
9. System validates the request against conflicts and policy rules.
10. If valid, booking is confirmed and stored.
11. User and admin can view booking details and history.

8. Admin Features

The admin dashboard should provide the following capabilities:

- create, edit, and deactivate user accounts,
- add, update, and remove rooms,
- view all upcoming and past bookings,
- cancel or modify bookings when required,
- monitor utilization and booking conflicts,

- maintain access permissions for different cores or groups.

9. User Features

Authorized users should be able to:

- log in securely,
- search rooms by block, capacity, purpose, and time,
- view availability of rooms,
- create bookings for permitted purposes,
- view their own booking history,
- cancel or edit their own bookings if allowed by policy.

10. Functional Requirements

Module	Requirement
Authentication	No self-registration; only admin-created accounts are allowed.
Access Control	Access should be restricted only to approved users, coordinators, or authorized cores.
Room Database	Maintain all block-wise room details including block name, room number, seating capacity, and availability status.
Booking Engine	Support booking based on date, time slot, purpose, and participant capacity requirements.
Conflict Handling	Prevent overlapping or duplicate bookings for the same room during the same time slot.
Purpose Validation	Ensure bookings are only made for valid purposes such as OA, Interviews, and PPTs.
Admin Control	Provide full visibility and management of users, rooms, permissions, and booking records.

11. Non-Functional Requirements

- Security: authentication, role-based access, and controlled admin permissions.
- Reliability: booking data should remain consistent and conflict-free.
- Scalability: the system should support more rooms and blocks in future.
- Usability: the interface should be simple and easy for OCS coordinators to use.
- Maintainability: room data and user permissions should be easy to update.

12. Suggested Database Entities

The implementation may include the following core entities:

- Users: stores admin-created accounts and roles.
- Blocks: stores block names such as A, B, C, CSE, LHC, etc.

- Rooms: stores room details, capacity, and block mapping.
- Bookings: stores date, time, purpose, and status of each booking.
- Permissions: stores access rules for cores or user groups.

13. Suggested Validation Rules

- End time must be greater than start time.
- Booking duration should be positive and valid.
- Room capacity should not be exceeded.
- Booking purpose must be one of OA, Interviews, or PPTs.
- Duplicate overlapping bookings must be rejected.

14. Future Enhancements

- approval workflow for sensitive bookings,
- calendar view for all rooms,
- email or notification on booking confirmation,
- booking analytics and usage reports,
- exportable daily or weekly schedules,
- bulk import of blocks and room details.

15. Final Task Statement

Build a secure OCS IITH Room Booking System that allows only admin-created and authorized users to book rooms across multiple IIT Hyderabad blocks for OA, Interviews, and PPTs. The system must enforce proper authentication, prevent overlapping bookings, respect room capacity, and support future expansion for all blocks and rooms in IIT Hyderabad.